# Rama Rao Duggirala

Security & Network Engineer
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## Professional Summary

Security and Network Engineer with 2 years of hands-on experience securing production infrastructure: Fortinet FortiGate 200G NGFW, network segmentation, firewall policy management, Splunk SIEM monitoring, traffic analysis (Wireshark, Nmap), VAPT, and incident response across RHEL and PostgreSQL environments. Certified Ethical Hacker (CEH) applying Zero Trust and least-privilege principles.

## Experience

**Security / Network Engineer** June 2025 – Present
*Quantum AI Global* *Hyderabad, Telangana*

- Cut inter-zone attack surface [X]% by segmenting the network into 5 FortiGate 200G zones (DMZ, Application, Database, LAN, Management) with least-privilege policies, NAT, and VLANs.
- Trimmed the firewall ruleset from [X] to [Y] policies through rule reviews; enforced IPS/IDS, web filtering, and application control on the remainder.
- Cut MTTD from [X] to [Y] min and MTTR to [Z] hrs by onboarding [N] log sources into Splunk SIEM, writing [N] SPL correlation searches mapped to MITRE ATT&CK, and running a severity-based triage workflow.
- Detected [X] port scans and lateral-movement attempts in [timeframe] using Wireshark and Network Behavior Analysis (NBA) on SPAN feeds from [N] VLANs; tuning cut false positives from [X]% to [Y]%.
- Sustained [99.X]% perimeter uptime with FortiGate High Availability (HA) and SSL VPN / IPsec access for [N] remote users, with zero exposed management ports.
- Closed [X]% of critical vulnerabilities within [Y] days using [Nessus / OpenVAS] scans prioritized by exploitability and asset criticality.
- Hardened [N] RHEL 9.5 servers to [CIS Benchmark] ([X]% compliant) and enforced RBAC with JWT authentication on [N] applications, with zero unauthorized-access incidents.
- Restored a production server from an initramfs/GRUB boot failure in [X] hrs with zero data loss.

**Software Developer Intern** Sept 2024 – June 2025
*Quantum AI Global* *Hyderabad, Telangana*

- Found [N] vulnerabilities ([X] critical/high) across web and network assets using Nmap, Metasploit, and Wireshark; [Y]% remediated by the end of the internship.
- Moved [X]% of traffic on [N] FastAPI/Django services to HTTPS/TLS after finding unencrypted endpoints.

## Projects

**Enterprise Network Security Architecture** | *FortiGate 200G, Splunk, NBA, DLP, DAM*

- Designed a layered defense-in-depth architecture (Prevent → Protect → Detect → Correlate → Respond → Recover → Improve) integrating FortiGate, Splunk SIEM, NBA, DLP, Database Activity Monitoring (DAM), and DDoS protection.
- Defined DLP and DAM monitoring requirements for [N] data stores: sensitive-data movement, privileged database access, unusual queries, bulk extraction, and schema changes.
- Established KPIs (MTTD, MTTR, critical-vulnerability aging, log-source coverage, false-positive rate) and governance for policy reviews, detection tuning, backups, and incident-response exercises.

**Secure Network Traffic Analyzer** | *Wireshark, Python, Nmap*

- Built a Wireshark and Python workflow to capture, decode, and flag suspicious traffic on monitored LAN interfaces.
- Mapped hosts and open ports with Nmap to identify exposed or misconfigured services.

**Text Encryptor** | *Python, AES, RSA, SHA, DES*

- Implemented symmetric and asymmetric encryption (AES, DES, RSA) and SHA hashing for secure data handling and integrity verification.

## Technical Skills

**Firewall & Network Security**: Fortinet FortiGate 200G (NGFW), Stateful Firewall, NAT, Routing, VPN (IPsec, SSL), IPS/IDS, Application Control, Web Filtering, Network Segmentation, VLANs, SPAN/Port Mirroring
**Monitoring & Detection**: Splunk SIEM, Log Correlation, Alert Triage, Threat Hunting, Network Behavior Analysis (NBA), Anomaly Detection, Detection Tuning, MTTD/MTTR
**Vulnerability & Testing**: VAPT, Nmap, Wireshark, Metasploit, Vulnerability Assessment, Risk Prioritization, Remediation and Retesting
**Data Security**: Trellix DLP, Database Activity Monitoring (DAM), Data Classification
**Linux & Infrastructure**: RHEL 9.5, Linux Administration, Apache, PostgreSQL, GRUB Recovery, Secure Production Deployment, TP-Link TL-SG3452, SONiC OS
**Protocols & Access Control**: TCP/IP, DNS, DHCP, IPv4/IPv6, Subnetting (CIDR/VLSM), HTTP/HTTPS, SSH, IPsec, SSL/TLS, RBAC, JWT
**Frameworks & Governance**: Zero Trust, Defense-in-Depth, Least Privilege, Security Policy Governance, Incident Response
**Programming**: Python, FastAPI, Django, JavaScript, REST APIs

## Certifications

Certified Ethical Hacker (CEH), EC-Council
Ethical Hacking & Penetration Testing on Web Applications; Networking; Python Django Development, Infosys Springboard
Cloud Computing, NPTEL | Cybersecurity, Skill Vertex

## Education

**Chalapathi Institute of Engineering and Technology (Autonomous)** 2021 – 2025
*B.Tech in Computer Science & Engineering (Data Science)* *CGPA: 7.3*